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Solute Movement through Quartz–Diorite Saprolite Containing Quartz Veins and Biological Macropores

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ABSTRACT

Saprolite is not permitted for use in on-site wastewater disposal in North Carolina if quartz veins occur within 60 cm (2 ft) of the septic trench bottom because such features might transmit raw sewage quickly to groundwater. This study evaluated the time of travel (TOT) of a Br^- solute through quartz veins in saprolite. Two 150 by 150 cm drainfields were constructed over saprolite containing quartz veins. Following saturation of the saprolite, a Br^- tracer and dye were applied for a specific time period. The drainfields were then excavated to 90 cm, the dye pattern mapped, and soil samples collected for Br^- analysis. There was no significant difference ($\alpha = 0.10$) in depth of Br^- penetration in saprolite with and without quartz veins. On the other hand, mean depth of Br^- penetration increased from 14 cm for saprolite without macropores to 40 cm in saprolite with macropores (root channels) that extended to depths between 40 to 90 cm. A simple time of travel model predicted maximum depth of solute movement accurately where macropores were not present. To improve predictions for macropores, effective saturated hydraulic conductivities for saprolite with and without macropores were calculated and then used to estimate different rates of solute movement through saprolite when macropores were and were not present.

SAPROLITE is bedrock that has weathered in place to the point that it is porous and can be dug with a spade (Becker, 1895). Saprolite occurs throughout the Piedmont and Mountains of the southeastern USA (Daniels et al., 1984). As the land available for conventional septic systems decreases, wastewater disposal in saprolite is gaining in popularity in some states. Health regulations will generally not permit its use if the saprolite contains fractures or veins. Fractures are simple planes of weakness or cracks in saprolite. Veins are tabular or plane-shaped bodies that contain minerals that differ from those in the surrounding saprolite (Guilbert and Park, 1986).

Most saprolites contain geologic fractures and gravel-filled veins that *appear* capable of conducting water rapidly. When fractures and veins are present in saprolite, it has been assumed that they are the primary voids for conducting water (Pavich, 1986). The lack of conclusive scientific data on water movement through quartz veins has caused public health officials in North Carolina to prohibit installation of septic system drainfields in saprolites that contain veins or fractures. Any site will be declared unsuitable for waste disposal if veins or fractures are present within 60 cm (2 ft) of the trench bottom (NCDEHNR, 1991).

Williams et al. (1994) used core samples (7.6 cm in diam. and 7.6 cm in height) to determine hydraulic conductivities of quartz–phyllite saprolite with and without quartz veins. There was no statistical difference in the geometric mean hydraulic conductivities between populations. It is not known whether the results from this study apply to longer and wider veins that also occur in saprolite. The results do suggest that quartz veins are not causing by-pass flow to occur in saprolite.

In soils that are suitable for waste disposal, effluent moves through the soil relatively slowly so that there is ample time for bacteria and viruses to be removed from the effluent by filtration or adsorption onto particle surfaces. Bouma et al. (1972) proposed that for a sandy loam soil, adequate purification would occur if effluent traveled downward a distance of 60 cm in 24 h or more. This suggests that time of travel computations could provide a way of estimating a soil's waste purifying potential, which is difficult to assess exactly, by using soil properties which are more easily measured.

The TOT for solutes to move specific distances (L_s) in soil or saprolite by convection or mass flow can be estimated from the average pore water velocity ($\bar{V}$) where:

$$\mathrm{TOT} = L_s/\bar{V} \qquad [1]$$

The average pore water velocity is determined from Darcy's Law as:

$$\bar{V} = [-K(\theta)\ (\Delta H/L_w)]/\theta \qquad [2]$$

where θ is the volumetric water content, $K(\theta)$ is the hydraulic conductivity at the water content θ, and $\Delta H/L_w$ is the hydraulic gradient where ΔH is the difference in total soil water pressure head between two points that are a distance L_w apart. Equations [1] and [2] can be combined to determine solute travel times from Darcy's Law as:

$$\mathrm{TOT} = -L_s\ \theta/[K(\theta)\ \Delta H/L_w] \qquad [3]$$

Because $K(\theta)$ is an average rate of water movement over an area, the above TOT model may perform poorly when there are spatial variations in hydraulic conductivity due to macropores (or quartz veins) rapidly conducting solute faster than the soil or saprolite matrix. The TOT model might be improved by using separate K values to account for flow along macropores and matrix. These effective hydraulic conductivity values could be determined experimentally by applying solute to the saprolite for a specific time period, and then observing how deep the solute traveled through the matrix and macropores separately. With such data the effective hydraulic conductivity could be calculated by the equation:

$$K_e(\theta) = -L_s\ \theta/[T\ \Delta H/L_w] \qquad [4]$$

where $K_e(\theta)$ is the effective hydraulic conductivity of the macropore or matrix, L_s is the depth the solute has moved in a vertical profile, T is the time that the solute was applied to the profile.

The objectives of this study were to: (i) evaluate water

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Abbreviations: $K(\theta)$, hydraulic conductivity when volumetric water content is θ; ΔH difference in hydraulic head; L_s distance of travel of solute; L_w, distance used for hydraulic gradient; TOT, time of travel.

and solute flow along quartz veins in a quartz-diorite saprolite, (ii) evaluate the accuracy of a time of travel model that predicts depth of solute movement under field conditions, and (iii) estimate effective hydraulic conductivity of saprolite containing quartz veins using a time of travel model.

MATERIALS AND METHODS

Preliminary Site Characterization

The Piedmont study site was located in a metamorphosed quartz-diorite saprolite in Randolph County, North Carolina, near the town of Ramseur (Carpenter, 1982). The soil at the study site was classified as Vance (clayey, mixed, thermic Typic Hapludult). A truck-mounted soil probe, 7.5 cm in diam., was used to collect soil and saprolite samples to a depth of 3.5 m from seven separate core locations that were selected randomly within a 12 by 12 m area. The extracted samples were cut to 6 cm lengths, and each core section was encased in wax and its saturated hydraulic conductivity determined (Amoozegar, 1988). The hydraulic conductivity with depth was graphed to determine the upper boundary of the C horizon where the permeable saprolite normally occurs (Schoeneberger and Amoozegar, 1990). A narrow trench was then excavated within the 12 by 12 m sampling area to identify quartz veins and to describe the soil-saprolite profile. Bulk samples and 10 Uhland cores (7.6 cm diam. by 7.6 cm in length) were collected from the C horizon. Bulk samples were oven dried (105°C), and crushed to pass a 2-mm mesh sieve. Particle-size distribution was determined using the pipette method (Gee and Bauder, 1986). Bulk density was determined after Uhland cores were oven dried (105°C).

A large pit (11 by 6 m) was excavated to a depth of ≈ 2 m to reach the Cl horizon and to expose quartz veins. The pit bottom contained three main quartz veins (Fig. 1) that ranged from 1 to 120 mm in width and were continuous for horizontal distances that exceeded 11 m. Many smaller veins were present both parallel and perpendicular to the main veins.

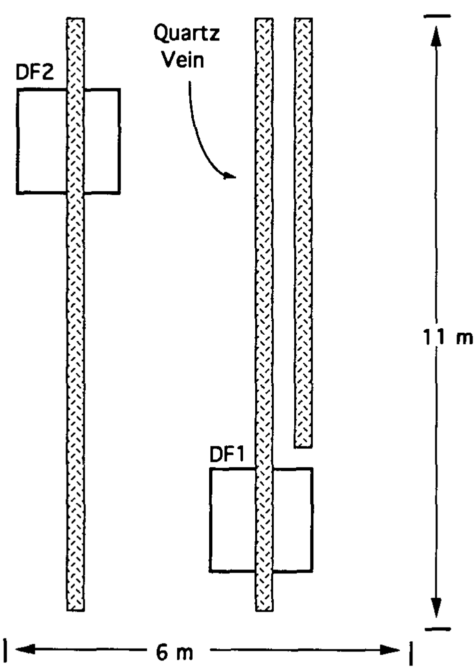

Fig. 1. Plan view of study site showing the quartz veins studied and locations of the two drainfields (DF1 and DF2). The diagram is idealized as veins varied in width and in orientation.

Drainfield Construction

The base of the pit was carefully cleared and leveled by hand using shovels. Two watertight dikes (150 by 150 cm) made of commercial plywood were assembled and placed over the quartz veins (Fig. 1). Dike sidewalls were buried to a depth of 15 cm leaving 15 cm above the surface. After installation, the outside edge of the dike was backfilled with bentonite pellets, and capped by a thin layer of concrete to shed rainwater away from the dike. Within the diked drainfield, loose saprolite within 8 cm of the dike walls was compacted to ensure that sidewall leakage would not occur.

Four tensiometers were installed in each drainfield to determine hydraulic gradients (Fig. 2). Two tensiometers were installed at each of two depths (30 and 60 cm) near a vein, and two were installed away from the vein at the same depths using standard techniques (Cassel and Klute, 1986). Soil water pressure head was measured by pressure transducer (Tensimeter, Soil Measurement Systems, Tucson, AZ)[1]. Tensiometers away from the vein appeared to be in the saprolite matrix that was not affected by the vein.

Two neutron access tubes were installed to a depth of 60 cm in each drainfield (Fig. 2). The tubes were sealed at the drainfield surface with 2 cm of bentonite. A calibration curve was developed outside the drainfields to determine volumetric water content (θ) using standard methods (Gardner, 1986).

A 5-cm thick layer of coarse sand was placed on the instrumented drainfield to protect the surface. A portable shelter was installed to protect drainfields from rainfall.

Hydraulic Conductivity

Hydraulic conductivity was measured in situ for each drainfield. Water application was controlled by a float valve attached to the sidewall of the drainfield and connected by hose

[1] The use of trade names in this publication does not imply endorsement by the North Carolina Agricultural Research Service of the product named, nor criticism of similar ones not mentioned.

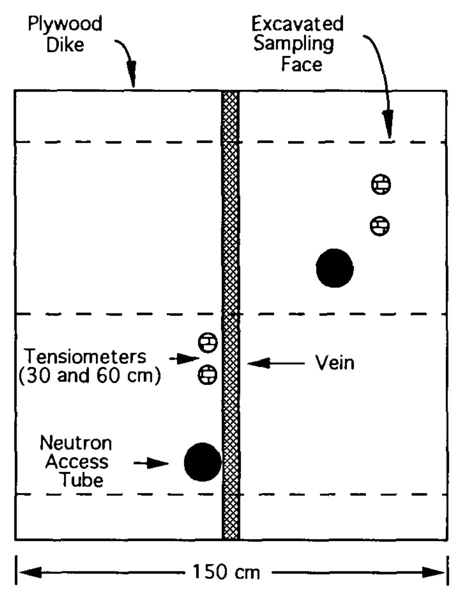

Fig. 2. Plan view of a diked drainfield showing dimensions and instrumentation. Dashed lines represent sampling faces. Tensiometers were installed in duplicate at depths of 30 and 60 cm. Access tubes were installed to 60 cm.

to a mobile 1900 L tank. The float valve maintained a 2-cm head of water above the sand-covered surface in the drainfield. When flooded, the drainfield was covered tightly by plastic film to prevent evaporation. After tensiometers at the 30-cm depth had indicated that the soil water pressure head was approximately 0 cm and that steady-state conditions had been reached, the volume of water infiltrating into the drainfield during a 24-h period was carefully measured and hydraulic conductivity was determined by using the following form of Darcy's equation:

$$K = -Q/[A\ T\ \Delta H/L_w] \qquad [5]$$

where K is equal to the hydraulic conductivity, Q is the volume of water infiltrating the drainfield, A is the drainfield area (2.25 m^2), $\Delta H/L_w$ is the hydraulic gradient between the depths of 0 and 30 cm, and T is time.

Application of Dye and Bromide

After completion of K measurements, a 0.1% (by weight) dye solution of Acid Blue 29 was added to the drainfield water. The dye was applied for the duration of the experiment to stain water-conducting macropores.

After a hydraulic gradient of approximately -1 cm cm^{-1} was attained by the flooding regime, a 1000 mg L^{-1} solution of Br^- (as KBr) was added to the dye solution in the drainfields for the time determined using Eq. [3] for each drainfield. For Eq. [3], L_s was chosen to be 30 cm, θ was measured by neutron probe at the 30 cm depth, $K(\theta)$ was determined as described previously, and $\Delta H/L_w$ was the hydraulic gradient between the depths of 0 and 30 cm. When the desired time had elapsed, excess water was drained from the surface, and the shelter, plumbing and instruments were removed.

Soil samples for Br^- analysis were collected from three vertical faces that were exposed in each drainfield. The sampling faces were ≈40 cm apart, 90 cm deep, and perpendicular to the quartz veins (Fig. 2). Each face was excavated by backhoe to a depth of 90 cm and smoothed by knife and shovel. A sampling grid (120 by 90 cm) with 10 by 10 cm square cells was then placed against the exposed face. Approximately 100 g of moist soil was collected by knife from the center of each grid cell and placed in an air-tight soil moisture can. A total of 648 samples were collected from the two drainfields for Br^- analysis.

Each soil sample was oven dried (105°C) to determine water content. The soil was then crushed, mixed, and a 10-g subsample collected. Fifty mL of distilled water were added to each subsample and the suspension was shaken for 15 min. Bromide concentration was determined by a commercial ion analyzer (Orion Research Microprocessor Ionalyzer Model 901, Cambridge, MA) with a Br^- electrode that detected Br^- in concentrations as low as 0.04 mg L^{-1}. The Br^- concentration was expressed as mg L^{-1} and graphed by sampling grid location for each of the six sampling faces (Fig. 3).

Depth of Br^- Penetration and Effective Hydraulic Conductivity

The sampling cells from each profile face were grouped into *miniprofiles* to determine the maximum depths that Br^- moved downward through the quartz veins and saprolite matrix. A miniprofile was a section of the sampling face that consisted of one column of sampling cells. Each column contained nine vertical cells that extended from the drainfield surface to the base of the sampling grid at a depth of 90 cm (Fig. 3). Miniprofile dimensions were 10 by 90 cm. Across each sampling face there were 12 miniprofiles, and each drainfield contained 36 miniprofiles.

Miniprofile Number

Depth (cm)	1	2	3	4	5	6	7	8	9	10	11	12
10	75	26	222	351	271	58	97	30	45	17	156	601
20	170	812	463	539	29	24	4	5	6	16	363	999
30	999	878	273	569	17	4	[illegible]	12	6	38	10	346
40	29	501	743	324	24	3	4	139	310	249	434	130
50	4	77	403	25	8	1	1	3	1	2	112	67
60	26	10	16	32	11	13	14	16	17	25	643	954
70	1	2	1	3	1	1	1	1	3	3	80	188
80	1	7	4	1	8	1	1	1	1	1	1	1
90	4	9	16	18	[illegible]	2	2	2	2	2	2	2

Fig. 3. Typical drainfield face with Br^- concentrations (mg L^{-1}) identified in each cell. A miniprofile is represented by the hashed cells. Maximum depth of 100 mg L^{-1} Br^- in each miniprofile is given by the underlined concentration. Miniprofiles 5, 6, 7, and 8 contain a quartz vein, which is represented by the double lines.

The maximum depth of Br^- penetration was determined separately for each miniprofile. This maximum depth was defined as the lowest cell of a miniprofile that had a Br^- concentration ≥100 mg·L^{-1} (10% by weight of the 1000 mg Br^- L^{-1} added). This cutoff concentration was selected because Br^- concentrations <100 mg L^{-1} were close to background Br^- concentrations in some drainfield faces. The Br^- concentrations in each miniprofile were examined in order from the bottom-most cell to the top-most cell to identify the maximum depths. Figure 3 illustrates maximum depths of Br^- penetration for 12 miniprofiles that were defined using this procedure.

To evaluate the accuracy of Eq. [3], the maximum depths of Br^- penetration found for each miniprofile were compared with target values. For this study, the model's prediction was considered *successful* if maximum depth of Br^- penetration within a miniprofile was between 10 and 40 cm. Because Br^- samples were collected from a square 10 by 10 cm grid cell, the samples collected from what we term *the 40-cm depth* represented Br^- that was actually collected between the depths of 30 to 40 cm. The model prediction was considered to have failed when maximum depth of penetration was >40 cm within a miniprofile.

The effective hydraulic conductivity, K_e, of each miniprofile was calculated with Eq. [4]. For each miniprofile, L_s was equal to the observed maximum depth of Br^- penetration found for that miniprofile, θ was the water content determined by neutron probe, T was the time interval over which the Br^- was applied, and $\Delta H/L_w$ was computed for depths of 0 and 30 cm.

The maximum depth of Br^- penetration and K_e values were compared among the miniprofiles using analysis of variance procedures (SAS Inst., 1985). Each drainfield was considered to be one experimental replicate. Mean values for maximum depth of Br^- penetration and Ke were compared. In the first comparison separate means were determined for miniprofiles with quartz veins and without quartz veins to evaluate whether solute moved along veins faster than through the matrix. If a miniprofile contained a quartz vein in any of its nine cells, it was classified as a miniprofile containing a quartz vein. Mean values were also compared between miniprofiles with and without macropores. A miniprofile was considered to contain a macropore if blue dye was observed in a cylindrical channel at a depth >40 cm below the drainfield surface at the time of sampling. The presence of dyed channels above 40 cm did not affect our classification of the miniprofile, because the channels could not directly transmit Br^- to depths >40 cm.

Table 1. Bulk density and particle-size distribution of the quartz–diorite saprolite's matrix in the C1 horizon. Bulk density values were determined from 10 core samples.

	Bulk density	Sand (2–0.05 mm)	Silt (0.05–0.002 mm)	Clay (<0.002 mm)
	g cm^{-3}		%	
Mean	1.39	43.05	37.10	19.85
Range	0.19	23.05	3.35	21.33

RESULTS

The saprolite matrix had a loam textural class (Table 1). Saturated hydraulic conductivity measured on waxed cores decreased to 0.01 cm h^{-1} in the Bt horizon, but was >4 cm h^{-1} in the C horizons (Fig. 4). The drainfields were installed at a depth of 1.95 m, near the upper boundary of the C1 horizon where hydraulic conductivity increased with depth. However, the slowly permeable BC horizon had a diffuse, wavy lower boundary, and each drainfield included portions of the BC horizon. Shallow excavations were made around each drainfield during the experiment to determine if lateral flow was occurring. No saprolite containing dye was seen outside the diked enclosure. This indicated that substantial lateral flow probably had not occurred. Hydraulic conductivities for the drainfields are reported in Table 2. The hydraulic gradients were near unity, and soil water pressure heads were near 0 cm or greater for both drainfields at a depth of 30 cm.

Quartz veins that were exposed in drainfield faces ranged from 1 to 12 cm in width. Most quartz veins were 2 cm wide, and most quartz gravels in the veins were 2 cm in diam. Gravel in veins appeared to be coated with clay. Some quartz veins formed an angle of 90° to the horizontal surface. Most quartz veins examined in the drainfield exposures, however, dipped between 50 and 80° from the horizontal. Four of the six drainfield faces had multiple quartz veins. Five of these faces had at least one vein that extended to the surface of the drainfield. Veins that were continuous to the surface extended to depths between 50 to 90 cm or more.

The frequency distributions for maximum depth of Br^- penetration in miniprofiles with and without quartz veins are shown in Fig. 5A. Graphical analysis showed that the frequency distributions for both groups were log-normal. Geometric means for maximum depths of Br^- penetration are listed in Table 3. Mean depths of Br^- penetration in miniprofiles with and without quartz veins were not

Table 2. Values for water content soil water head, hydraulic gradient (0- to 30-cm depth) and hydraulic conductivity for both drainfields. The *K* values were computed using Eq. [5]. Values for water content and soil water heads are means of data collected at the 30-cm depth.

Drainfield	Water content	Soil water pressure head	Hydraulic† gradient	Hydraulic conductivity
	cm^3 cm^{-3}	cm	cm cm^{-1}	cm h^{-1}
1	0.42	+13	−0.7	0.33
2	0.43	−2	−1.1	0.08

† Difference in total soil water pressure head between the saprolite surface and a depth of 30 cm.

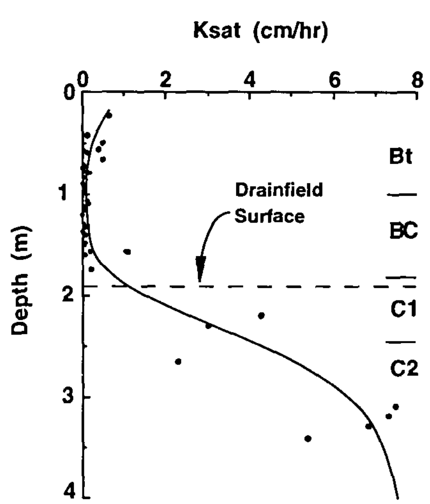

Fig. 4. Saturated hydraulic conductivity with depth as determined from undisturbed waxed cores (7.6 cm diam. by 6 cm in height). Drainfields were placed at a depth of 1.95 m at the top of the C1 horizon.

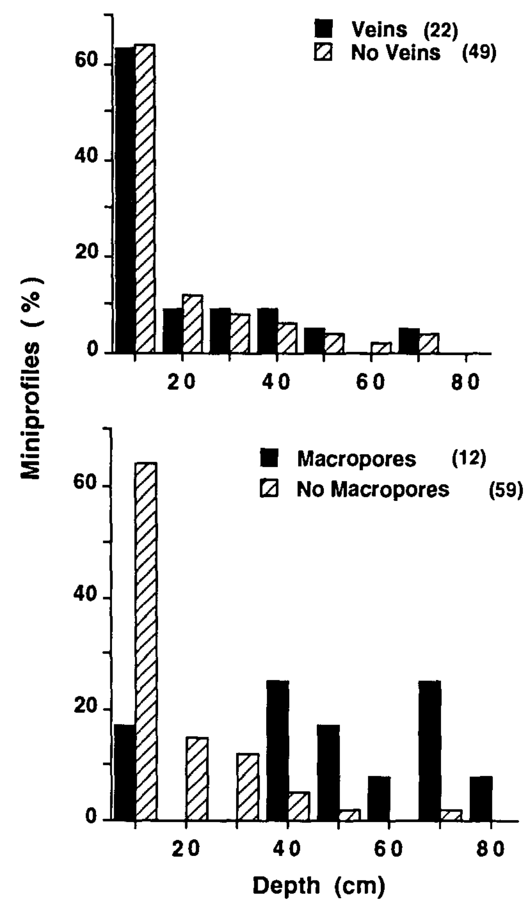

Fig. 5. Frequency distribution of maximum depth of Br^- penetration in miniprofiles: (*A*) with and without quartz veins, and (*B*) with and without macropores.

Table 3. Mean values for maximum depths of Br^- penetration and effective hydraulic conductivity (K_e) in miniprofiles with and without quartz veins, and miniprofiles with and without macropores. The proportion of miniprofiles having maximum depths of Br^- penetration between 10 and 40 cm are also shown as an indicator of the success of Eq. [3] in predicting solute transport.

Miniprofile group†	Maximum depths of Br^- penetration‡	Miniprofiles with penetration depth between 10 and 40 cm	Effective hydraulic conductivity
	cm	Prop. of Profiles	cm h^{-1}
Comparison 1			
Veins (22)	21	0.87	0.16
No Veins (50)	26	0.86	0.17
Difference	NS	–	NS
Comparison 2			
Macropores (13)	40	0.46	0.28
No Macropores (59)	14	0.95	0.10
Difference	*	–	*

* Significantly different at $\alpha = 0.05$ determined by F test, NS, not significantly different, and a dash indicates a comparison was not applicable.
† Number in parentheses is number of miniprofiles in that miniprofile group.
‡ *Maximum depth* refers to the lowest cell in a miniprofile that had a Br^- concentration ≥ 100 mg L^{-1} in the saprolite solution.

significantly different ($\alpha = 0.05$). This indicated that Br^- moved downward through the vein at a rate similar to that of the saprolite matrix. The veins were stained by the dye within a depth of 10 cm of the surface, which indicated that water did move along the veins for a short distance near the surface of the drainfield.

The macropores observed were cylindrical pores, 1 to 5 mm in diam. that often contained roots. Water-conducting macropores were identified by dye staining. The frequency distribution for maximum depths of Br^- penetration in miniprofiles with dye-stained macropores differed from those noted previously (Fig. 5B). Where macropores were not present, the data appeared to conform to a log-normal distribution. The mean depth of maximum Br^- penetration was 40 cm where macropores were present at depths ≥ 40 cm and only 14 cm where they were absent (Table 3). This difference in depth of Br^- penetration was statistically significant ($\alpha = 0.05$).

Evaluation of the time of travel model is shown in Table 3. In miniprofiles with and without quartz veins, the proportion of miniprofiles with maximum depths of Br^- penetration between 10 and 40 cm was 0.86 or 0.87. When no macropores were present at depths ≥ 40 cm, the proportion of miniprofiles with a maximum depth of Br^- penetration between 10 and 40 cm was >0.95; but when macropores were present the proportion dropped to 0.46. In other words, for the 13 miniprofiles that contained macropores at depths ≥ 40 cm, only six of them had maximum depths of Br^- penetration that were <40 cm.

The effective hydraulic conductivities (K_e) calculated using Eq. [4] were not affected by quartz veins (Table 3). When macropores were present, the mean K_e of 0.28 cm h^{-1} was significantly greater ($\alpha = 0.10$) than that found for the saprolite matrix without macropores, which had a K_e of 0.10 cm h^{-1}.

Time of travel estimates made using measured and effective $K(\theta)$ values are compared in Table 4. Two predicted values are shown, one based on drainfield 1 ($K(\theta)$ of 0.33 cm h^{-1}) and the other on drainfield 2 ($K(\theta)$ of 0.08 cm h^{-1}). A mean predicted TOT based on the geometric mean of these $K(\theta)$ was 89 h. This was very close to the computed effective TOT of 86 h shown in Table 4. Effective TOTs showed greater differences when travel through saprolite with and without macropores were compared.

Table 4. Comparison of predicted and effective time of travel estimates. Predicted values were computed using Eq. [3] and drainfield data shown in Table 2. Effective values were computed based on the effective K values shown in Table 3 and means of the water contents and hydraulic gradients shown in the table.

Saprolite miniprofiles	Time of travel (30 cm) Predicted†	Effective
	h	
All	5–147	86‡
With macropores	NA	51
Without macropores	NA	142

† Measured $K(\theta)$ were 0.33 and 0.08 cm h^{-1} for Drainfields 1 and 2, respectively. NA is not applicable.
‡ Determined using an effective K of 0.165 cm h^{-1} as determined from Comparison 1 in Table 2.

DISCUSSION

The results of this study have shown that quartz veins that were 1 to 12 cm in width and occurred in the C1 horizon of saprolite did not conduct water significantly faster than the saprolite matrix. These veins posed no apparent hazard to on-site wastewater disposal. These results should be considered site-dependent. Previous work by Guertal (1992), however, has shown that the water-conducting pores of C1 horizons are frequently either coated or partially plugged with Fe oxides or clay. Such coatings were observed on the gravels of the veins we studied. We'do not know whether veins that are deeper in the saprolite would conduct water and solutes at rates comparable to those of the saprolite matrix. It is possible that deeper veins would have less coatings of clay and therefore would conduct water and solutes at rates greater than those measured here.

For this study we applied solute over the time period that was sufficient for it to travel to a depth of 30 cm as predicted by Eq. [3]. We considered the model's prediction to be successful if the observed maximum depth of Br^- penetration was between 10 and 40 cm. Failure occurred if the maximum depth of Br^- penetration exceeded 40 cm, because for waste disposal purposes adequate purification of effluent does not occur if it travels too quickly. As shown in Table 3, Eq. [3] worked well except where macropores extended from the surface to depths ≥ 40 cm. As shown in Fig. 5, the maximum depth for Br^- penetration was 10 cm for most miniprofiles except when macropores extended to depths ≥ 40 cm. Thus, Eq. [3] tended to underestimate the time needed for penetration to occur in most of the miniprofiles examined.

Effective hydraulic conductivities allowed estimating separate TOT values for saprolite with and without macropores. This had the advantage of indicating that there will be at least two different rates of solute movement, with the slower rate occurring through saprolite matrix without macropores. Even where macropores extended to depths

$\geq$40 cm, solute travel was still computed to be relatively slow requiring 51 h to move 30 cm. This is probably due to the macropores terminating in saprolite matrix at depths between 40 and 90 cm. While macropores would fill with solute solution, the solution did not penetrate into the surrounding matrix quickly.

A complete site evaluation for wastewater treatment and disposal must consider factors such as water table depth, slope, depth to bedrock, and depth of the septic drainfield. If this saprolite were used for on-site wastewater disposal, preferential flow of water and solutes would occur along biological macropores if an organic clogging mat was not present. Biological macropores are probably found in most (if not all) operating septic drainfields and are not considered to be a problem. Future work in other saprolites must address larger quartz veins and veins without plugging materials.

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